Weather Data Analysis

Project 3 — Multi-Domain Data Analysis Portfolio

Executive Summary

This project analyzes a simulated daily weather dataset containing 2,000 observations from January 1, 2020 to June 22, 2025. The analysis evaluates temperature trends, rainfall patterns, seasonal variation, weather-condition frequency, extreme weather events, descriptive statistics, and correlations between key weather variables.

The dataset shows a relatively stable average temperature of 27.22°C, with temperatures ranging from 10.79°C to 45.00°C. Monsoon records the highest seasonal rainfall at 2,277.00 mm, while clear conditions account for 67.4% of all observations. Forty-eight extreme-weather records were identified using predefined thresholds.

Key Performance Indicators

Metric	Result
Observations	2,000
Date range	2020-01-01 to 2025-06-22
Average temperature	27.22°C
Average rainfall	3.76 mm
Average humidity	64.85%
Average wind speed	14.75 km/h
Extreme-weather records	48
Clear-weather days	1,348 (67.4%)

1. Dataset & Methodology

The dataset was generated for analytical demonstration and includes Date, Temperature, Humidity, Wind Speed, Rainfall, and Weather Condition fields. Additional Year, Month, Month_Name, and Season fields were derived from the date.

Season definitions used in the analysis were Winter (December–February), Summer (March–May), Monsoon (June–September), and Post-Monsoon (October–November).

2. Temperature Analysis

The mean and median temperature are both approximately 27.22°C. Monthly average temperatures remain within a relatively narrow range: June has the highest monthly average at 27.68°C, while September has the lowest at 26.79°C. The full dataset ranges from 10.79°C to 45.00°C.

3. Rainfall Analysis

Average daily rainfall is 3.76 mm and the median is 2.63 mm. April records the highest monthly rainfall total at 744.44 mm, while November records the lowest at 510.07 mm. The Monsoon season has the highest seasonal rainfall total at 2,277.00 mm.

4. Seasonal Patterns

Season	Avg Temp	Total Rainfall	Avg Humidity	Avg Wind
Winter	27.25°C	1,875.79 mm	63.89%	14.59 km/h
Summer	26.96°C	2,192.60 mm	65.35%	15.01 km/h
Monsoon	27.32°C	2,277.00 mm	65.43%	14.75 km/h
Post-Monsoon	27.46°C	1,181.65 mm	64.32%	14.57 km/h

5. Weather Condition Distribution

Clear conditions dominate the simulated observations, with 1,348 days (67.4%). Rainy conditions account for 494 days (24.7%), followed by Hot (83), Cool (43), and Heavy Rain (32).

Condition	Days	Share
Clear	1,348	67.4%
Rainy	494	24.7%
Hot	83	4.15%
Cool	43	2.15%
Heavy Rain	32	1.60%

6. Extreme Weather Events

Extreme events were defined using transparent thresholds: temperature ≥40°C for extreme heat, temperature ≤15°C for extreme cold, rainfall ≥20 mm for heavy rain, and wind speed ≥30 km/h for high wind. A total of 48 extreme-weather records were identified.

Event Type	Count
Extreme Heat	8
Extreme Cold	17
Heavy Rain	8
High Wind	15

7. Statistical Summary

Variable	Mean	Median	Minimum	Maximum
Temperature	27.22°C	27.22°C	10.79°C	45.00°C
Humidity	64.85%	65.00%	20.00%	100.00%
Wind Speed	14.75 km/h	14.82 km/h	2.00	33.68
Rainfall	3.76 mm	2.63 mm	0.00	26.33

8. Correlation Analysis

The major weather variables exhibit very weak linear relationships in this simulated dataset. Temperature versus rainfall has a correlation of 0.001, temperature versus humidity is -0.018, wind speed versus rainfall is 0.034, and humidity versus wind speed is 0.035. These results suggest that no strong linear relationship is visible among the variables under the simulated data-generating process.

9. Business & Operational Recommendations

1. Plan for seasonal rainfall: Increase preparedness during the Monsoon season and use rainfall history to support logistics and outdoor-operation planning.

2. Prepare for extreme weather: Establish contingency procedures for extreme heat, cold, heavy rainfall, and high winds, with operational adjustments when severe conditions are expected.

3. Optimize resource planning: Use weather trends to inform transportation, staffing, outdoor activities, and infrastructure maintenance.

4. Improve rainfall monitoring: Pay additional attention to higher-rainfall periods and maintain suitable drainage and water-management infrastructure.

5. Expand future analysis: Add air pressure, cloud cover, solar radiation, and location-specific variables and apply the workflow to real meteorological datasets for forecasting.

10. Data Quality Validation

Final validation confirmed 2,000 rows and 10 columns, with 0 missing values, 0 duplicate rows, and 2,000 unique dates. The date range is January 1, 2020 through June 22, 2025. Temperature ranges from 10.79°C to 45.00°C and rainfall ranges from 0.00 to 26.33 mm.

11. Conclusion

The project demonstrates an end-to-end weather data analysis workflow using Python, Pandas, NumPy, and Matplotlib. The analysis identifies stable temperature patterns, meaningful seasonal rainfall variation, predominantly clear conditions, and a measurable set of extreme-weather events.

Since the dataset is simulated, these results are intended for portfolio demonstration rather than real-world weather forecasting. The methodology can be transferred to reliable meteorological or government open-data sources for production analysis.

Report prepared as part of the Multi-Domain Data Analysis Portfolio.